(iii) Normal form:

$$\begin{aligned}\sqrt{3}x - y + 4 &= 0\\ (-\sqrt{3})x + y &= 4,\end{aligned} \tag{6.17}$$

Comparing the above equation with the equation $Ax + By + C = 0$.

Here $A = -\sqrt{3}$, and $B = 1$, $\sqrt{A^2 + B^2} = 2$

Therefore, dividing the above equation by 2, we get

$$\frac{-\sqrt{3}x}{2} + \frac{y}{2} = 2 \tag{6.18}$$

Comparing the above equation with the equation $x\cos\alpha + y\sin\alpha = p$

If we take

$$\begin{aligned}\cos\alpha &= \frac{-\sqrt{3}}{2} \text{ and } \sin\alpha = \frac{1}{2} \text{ and } p = 2\\ \Rightarrow \alpha &= 150^\circ = \frac{5\pi}{6} \text{ and length of the normal } (p) = 2\end{aligned}$$

The normal form is $x\ \cos\dfrac{5\pi}{6} + y\ \sin\dfrac{5\pi}{6} = 2$

To express the given equation to the required form, some times, it is more convenient to use property the proportionality of the coefficients of like terms.

Example 6.20 Rewrite $\sqrt{3}x + y + 4 = 0$ in to normal form.

Solution:

The required form $x\cos\alpha + y\sin\alpha = p$
Given form $-\sqrt{3}x - y = 4$ $(\because p$ is always positive$)$
Since both represent the same equation, the coefficients are proportional. We get,

$$\frac{\cos\alpha}{-\sqrt{3}} = \frac{\sin\alpha}{-1} = \frac{p}{4}$$

$$\frac{\cos\alpha}{-\sqrt{3}} = \frac{\sin\alpha}{-1} = \frac{p}{4} = \frac{\sqrt{\cos^2\alpha + \sin^2\alpha}}{\sqrt{\left(-\sqrt{3}\right)^2 + (-1)^2}} = \frac{1}{2}$$

(In componendo and dividendo, whenever a term is square off, then it should be square root off it.)

$$\cos\alpha = \frac{-\sqrt{3}}{2},\ \sin\alpha = \frac{-1}{2} \text{ and } p = \frac{4}{2}$$

$$\alpha = 210^\circ = \frac{7\pi}{6} \text{ and } p = 2$$

Normal form of the equation is

$$x\cos\frac{7\pi}{6} + y\sin\frac{7\pi}{6} = 2$$